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95

TRANSPORT PHENOMENA I, CHE 311

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A liquid film of constant thickness δ is flowing down an inclined flat conveyor belt of length L and width W that is moving with a constant velocity v_0 .

- 1- Develop a shell momentum balance to obtain the steady-state velocity profile in the liquid film. Assume L and W to be large compared to δ , so the disturbances at the edges of the system may be ignored.
- 2- Determine the viscous force of the liquid film on a unit area of the conveyor belt.
- 3- Determine the mass flow rate of the liquid film.

Your answers may be in terms of the following constant parameters:

μ and ρ : viscosity and density of the liquid

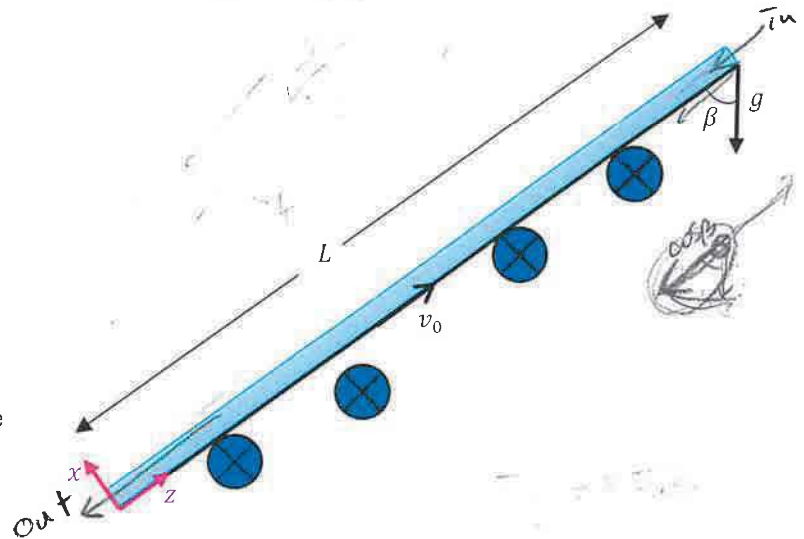
δ : thickness of the liquid film

L and W : length and width of the conveyor belt surface.

v_0 : velocity of the conveyor belt surface in z -direction.

g : gravitational constant.

β : angle between the conveyor belt surface and the direction of gravity.

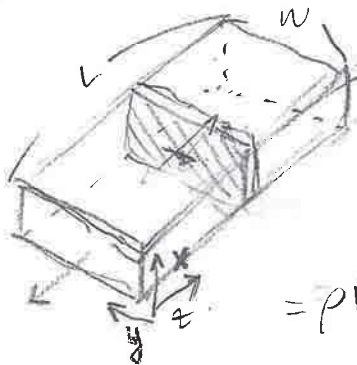


$$\int_V dV \frac{\partial \rho(v)}{\partial t} = - \int_S dS (\phi|n) + \int_V dV \rho(g)$$

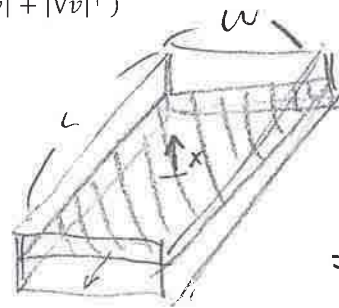
$$|\phi| = \rho(v)(v) + |\pi| = \rho(v)(v) + |\tau| + p|\delta|$$

$$|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$$

a) z momentum



$$\begin{aligned} \phi_{zz} &= \rho v_z v_z + p \delta_{zz} + \tau_{zz} \\ &= \rho v_z^2 + p_z + (-\mu) \left(2 \frac{dv_z}{dz} \right) \\ &= \rho v_z^2 + p_z \end{aligned}$$



$$\begin{aligned} \phi_{xz} &= \rho v_x v_z + p \delta_{xz} + \tau_{xz} \\ &= -\mu \left(\frac{dv_z}{dx} + \frac{dv_x}{dz} \right) \end{aligned}$$

$$W \Delta x (\phi_{zz}|_{z=L} - \phi_{zz}|_{z=0}) + L W (\phi_{xz}|_{x=x_0} - \phi_{xz}|_{x=x_0+\Delta x}) - (p g \cos \beta) (L W \Delta x) = 0$$

in the direction of z , the gravity should have negative sign

Name: Daahn Kim

$$\lim_{\Delta x \rightarrow 0} \frac{W \Delta x (\phi_{zz}|_{z=L} - \phi_{zz}|_{z=0}) + LW (\phi_{xz}|_{x=x_0} - \phi_{xz}|_{x=x_0+\Delta x}) - \rho g \cos \beta (LW \Delta x)}{LW \Delta x} = 0$$

$$\frac{(\phi_{zz})|_{z=L} - (\phi_{zz})|_{z=0}}{L} + \frac{d(\tau_{xz})}{dx} - \rho g \cos \beta = 0$$

(ρv_z^2 , P_z are constant at $z=L$ & $z=0$, so ϕ_{zz} is constant respect to z)

$$\frac{d(\tau_{xz})}{dx} = -\rho g \cos \beta$$

$$\tau_{xz} = (-\rho g \cos \beta)x + C_1, \text{ at } x=\delta, \tau_{xz}=0$$

$$0 = (-\rho g \cos \beta)\delta + C_1, \quad C_1 = \rho g \delta \cos \beta, \quad \tau_{xz} = (-\rho g \cos \beta)x + \rho g \delta \cos \beta$$

$$-\mu \frac{dv_z}{dx} = (-\rho g \cos \beta)x + \rho g \delta \cos \beta$$

$$\frac{dv_z}{dx} = \left(\frac{\rho g \cos \beta}{\mu} \right) x - \frac{\rho g \delta \cos \beta}{\mu}$$

$$v_z = \left(\frac{\rho g \cos \beta}{\mu} \right) \frac{x^2}{2} - \frac{\rho g \delta \cos \beta}{\mu} x + C_2$$

$$v_z = v_0 \text{ at } x=0, \quad v_0 = C_2$$

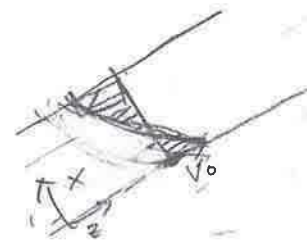
$$v_z = \left(\frac{\rho g \cos \beta}{\mu} \right) \frac{x^2}{2} - \frac{\rho g \delta \cos \beta}{\mu} x + v_0$$

(b) The viscous force of the liquid film on the surface of the conveyor belt

$$F_z = \int_0^L \int_0^W \tau_{xz}|_{x=0} dy dz = LW ((-\rho g \cos \beta) \cdot 0 + \rho g \delta \cos \beta) = LW \rho g \delta \cos \beta$$

The viscous force on a unit area

$$\frac{F_z}{A} = \frac{\int_0^L \int_0^W \tau_{xz}|_{x=0} dy dz}{\int_0^L \int_0^W dy dz} = \frac{LW \rho g \delta \cos \beta}{LW} = \boxed{\rho g \delta \cos \beta}$$



Name: Daohu Kim

(c)

mass flow rate $\dot{W}$: The liquid is flowing down, so to get mass flow rate, the velocity should have negative sign

$$\dot{W} = \frac{-\int_0^w \int_0^s v_z dx dy}{\int_0^w \int_0^s dx dy} = -\frac{1}{s} \int_0^s v_z dx$$

$$= -\frac{1}{s} \int_0^s \left(\frac{\rho g \cos \beta}{\mu} \right) \frac{x^2}{2} - \left(\frac{\rho g s \cos \beta}{\mu} \right) x + v_0 dx$$

$$= -\frac{1}{s} \left(\frac{\rho g \cos \beta}{\mu} \right) \left[\frac{x^3}{6} - s \frac{x^2}{2} + \left(\frac{\mu v_0}{\rho g \cos \beta} \right) x \right]_0^s$$

$$= -\frac{1}{s} \left(\frac{\rho g \cos \beta}{\mu} \right) \left(\left[\frac{s^3}{6} - \frac{s^3}{2} + \frac{\mu v_0}{\rho g \cos \beta} s \right] - 0 \right)$$

$$= - \left(-\frac{\rho g s^2 \cos \beta}{3\mu} + v_0 \right) = \boxed{\frac{\rho g s^2 \cos \beta}{3\mu} - v_0}$$